

Elias Hossain

CONTACT

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RESEARCH STATEMENT

I am a Ph.D. student at the University of Central Florida, advised by Dr. Niloofar Yousefi. My research is about **reliable large language models when humans cannot supervise at scale**. Four concrete questions drive the work. **(1)** Can pretrained transformers report calibrated uncertainty in their own attention? **(2)** Can we verify AI-generated supervision when the judges themselves share blind spots? **(3)** Can we monitor tool-using agents and intervene on unsafe trajectories, not just outputs? **(4)** Can we steer a model away from outdated parametric memory at test time, without retraining or retrieval? My applied vertical is **clinical and biomedical AI**, where the cost of confident errors is measured in patient harm, not benchmarks.

RESEARCH IMPACT

- **950+ citations, h-index 12, i10-index 12** (Google Scholar, July 2026).
- **12+ first-author peer-reviewed publications** in Q1 venues including *IEEE Transactions on Systems, Man, and Cybernetics: Systems, Computers in Biology and Medicine*, *Briefings in Bioinformatics*, and *Scientific Reports*.
- Most-cited lead-author work: a systematic review of NLP in electronic health records (**507 citations**), a university-deployed chatbot system (**64 citations**), and multimodal brain-tumor segmentation (**63 citations**).
- **5 public benchmarks** (74,590+ records) and **14+ open-source repositories** released to the community.
- **Peer reviewer for 7 international journals** (Nature Portfolio, Elsevier, PLOS, SAGE, Taylor & Francis, Springer) and the ICLR Trustworthy AI Workshop.

EDUCATION

University of Central Florida, College of Engineering and Computer Science, Orlando, FL, USA

Ph.D., Industrial Engineering & Management Systems

Aug 2025 – Present

- Computer Science concentration. Advisor: **Dr. Niloofar Yousefi**; Co-advisor: Dr. Ivan Garibay. Complex Adaptive Systems Laboratory.
- Dissertation: *Self-Verifying Scientific AI: A Verification-Centric Framework for Trustworthy Biomedical Reasoning*. Qualifying examination passed Summer 2026; expected graduation Spring 2028.

Mississippi State University, Mississippi State, MS, USA

M.S., Computer Science (Research), GPA 3.88/4.00

Jan 2024 – Jul 2025

- Thesis: *A Multi-Stage AI Framework for Topic Discovery in Scientific Abstracts*. Advisors: Dr. Andy D. Perkins, Dr. Shahram Rahimi.

RESEARCH EXPERIENCE

Graduate Research Assistant, Complex Adaptive Systems Laboratory, UCF Aug 2025 – Present

- Lead an independent program on trustworthy and safe LLMs: alignment under corrupted preference supervision, calibrated uncertainty for foundation models, runtime safety for tool-using agents, and inference-time control of frozen policies.
- Peer-reviewed acceptance at the **ICLR 2026 Trustworthy AI Workshop** and an invited poster at the **Amazon Trusted AI Symposium**. First-author manuscripts actively under review at NeurIPS and the ACL/EMNLP venues; published lead-author work appears in Q1 journals.
- Apply these methods to clinical decision support and antimicrobial-resistance discovery, including BIOGEN (multi-agent transcriptomic reasoning) and CareBench (process-aware evaluation of clinical agents).

- Built a contrastive-learning framework for anomaly detection in large-scale cybersecurity event logs, pairing contrastive sampling of rare malicious events with uncertainty estimation so that precision stays stable under heavy class imbalance. The work produced a peer-reviewed ICICT 2026 paper.
- Co-developed **BarkPlug V.2**, a retrieval-augmented chatbot deployed at MSU, with FAISS retrieval evaluated by RAGAS metrics and the System Usability Scale.

INDUSTRY
EXPERIENCE

Software Engineer, REVE Systems, Dhaka, Bangladesh Dec 2022 – Nov 2023
Deployed speech-recognition models (DeepSpeech, Wav2Vec 2.0) with 20% lower word error rate and 15% faster inference; delivered a voice-enabled extension for Bangladesh government e-governance services.

Senior AI/ML Researcher, Time Research & Innovation, UK Oct 2020 – Nov 2022
Deep-learning architectures for COVID-19 diagnosis from X-ray and CT imagery; co-developed a telemedicine platform for the UK healthcare system.

PEER-REVIEWED
PUBLICATIONS

Full list and live citation counts on [Google Scholar](#).

Journal Articles

Elias Hossain, Md. Mehedi Hasan Bhuiyan Nipu, Mohammad Sakib Mahmood, Md. Jakir Hossen, M. F. Mridha. "Safe and Scalable Collaboration in Multi-Agent LLM Systems: A Comprehensive Review." *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026. Accepted as a Regular Paper.

Elias Hossain, Rajib Rana, Niall Higgins, Jeffrey Soar, Prabal Datta Barua, Anthony R. Pisani, Kathryn Turner. "Natural Language Processing in Electronic Health Records in relation to health-care decision-making: A systematic review." *Computers in Biology and Medicine*, 2023. [\[link\]](#) **507 citations**.

Elias Hossain, Niloofar Yousefi. "Computational paradigms for antimicrobial resistance prediction: integrating multi-omics, structural modeling, and foundation artificial intelligence systems." *Briefings in Bioinformatics*, vol. 27, no. 3, 2026. [\[link\]](#)

Elias Hossain, Mehrdad Shoeibi, Ivan Garibay, Niloofar Yousefi. "BIOGEN: evidence-grounded multi-agent reasoning framework for transcriptomic interpretation in antimicrobial resistance." *Frontiers in Bioinformatics*, vol. 6, 2026. [\[link\]](#)

Elias Hossain, Tasfia Nuzhat, Shamsul Masum, Shahram Rahimi, Noorbakhsh Amiri Golilarz. "R-GAT: Graph Neural Networks for Cancer Document Classification." *Scientific Reports* (Nature Portfolio), 2026. [\[link\]](#)

Noorbakhsh Amiri Golilarz, **Elias Hossain**, Shahram Rahimi, Hossein Karimi. "A hybrid approach for pattern recognition and interpretation in age-related false memory." *Frontiers in Psychology*, 2025. [\[link\]](#)

Conference, Workshop & Symposium Papers

Elias Hossain, Maryam Rahimimovassagh, Subash Neupane, Mohammad Jahid Ibna Basher, Ivan Garibay, Niloofar Yousefi. "Fault-Tolerant Preference Alignment via Multi-Agent Verification." *ICLR 2026 Trustworthy AI Workshop*.

Elias Hossain, Maryam Rahimimovassagh, Niloofar Yousefi. "SAVe V.1: Multi-Agent Safe Alignment Verification for Preference-Based LLM Optimization." *Trusted AI Symposium*, Amazon JFK27, New York, NY, 2026. (Poster.)

Subash Neupane, **Elias Hossain**, Jason Keith, Himanshu Tripathi, Farbod Ghiasi, Noorbakhsh Amiri Golilarz, Amin Amirlatifi, Sudip Mittal, Shahram Rahimi. "From questions to insightful answers: Building an informed chatbot for university resources." *IEEE Frontiers in Education Conference (FIE)*, 2024. [link] **64 citations**.

Elias Hossain, Umesh Biswas, Charan Gudla, Sai Phani Parsa. "Learning Robust Representations for Malicious Content Detection via Contrastive Sampling and Uncertainty Estimation." *9th International Conference on Information and Computer Technologies (ICICT)*, USA, 2026. [arXiv]

Additional Peer-Reviewed Work

Seven Scopus-indexed Springer Nature book chapters (LNCS, LNNS, CCIS, LNEE, LNICST), and further refereed journal papers including multimodal brain-tumor segmentation (*Computers, Materials & Continua*, 2022; **63 citations**).

PREPRINTS & MANUSCRIPTS UNDER REVIEW

First-author unless noted.

"UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers." With S. R. Dipta, S. Neupane, R. Rana, R. Schwartz-Ziv, I. Garibay, N. Yousefi. [arXiv, project page] *ACL ARR meta-review: borderline conference accept; committed to EMNLP 2026*.

"ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems." With M. M. H. Nipu, F. T. J. Faria, T. N. Ornee, M. Sheikh. [arXiv, project page] Training-free information-bottleneck gates on every inter-agent channel; per-trace attribution shows undefended pipelines outsource safety to provider filters.

"NEXUS: Structured Runtime Safety for Tool-Using LLM Agents." With M. M. H. Nipu, T. N. Ornee, R. Rana, N. Yousefi. [arXiv, project page] Graded intervention policy over allow / block / confirm / revise; 27.3-point gain over rule-only selection at 0.205 ms median latency.

"Right Knowledge, Wrong Answer: Characterizing Parametric Temporal Conflict in Open-Weight Language Models." With S. Saha, T. N. Ornee, S. S. Jennifer, U. C. Biswas, S. R. Dipta, R. Rana, N. Yousefi. [arXiv, project page] Benchmark of 8,746 verified Wikidata transitions; activation patching localizes the outdated-fact preference to model-specific upper layers.

"Agreement Is Not Evidence: Measuring Correlated Failure in LLM Judge Consensus" (CorrFilter). Large-scale audit of whether LLM-judge banks fail independently, the assumption behind k-of-n consensus filtering for alignment data. Supervised by Prof. Ser-Nam Lim. [project page] *Under review*.

OPEN-SOURCE SOFTWARE & PUBLIC BENCHMARKS

All artifacts at huggingface.co/EliasHossain and github.com/eliashossain001.

Benchmarks. [nanobubbleeval](#) (51,566 records, schema-constrained biomedical extraction); [BioDivergence-Silver-v1.0](#) (11,900 claim pairs); [ptc-benchmark](#) (9,250 temporal-conflict QA); [carebench](#) (clinical agent trajectories); [CancerAbstracts](#) (1,874 abstracts).

Project pages (method figures, result tables, artifacts): [CareBench](#), [ChannelGuard](#), [NEXUS](#), [TAS](#), [UAT-LITE](#), and [CorrFilter](#). Research code is released at github.com/eliashossain001, including a 0.6B transformer trained from scratch in PyTorch and domain-adapted clinical language models.

SELECTED TALKS & PRESENTATIONS	Invited Talk , artificial intelligence in rural American healthcare, Mississippi Health Innovation Conference, Millsaps College, USA. [video] 2024
	Talk , “CITE V.1: Evidence-Grounded, LLM-Agentic Interpretation of RNA-Seq Clusters,” Molecular Machine Learning Conference (MoML), MIT Jameel Clinic, Cambridge, MA 2025
	Talk , “Fault-Tolerant Preference Alignment via Multi-Agent Verification,” Trustworthy AI Workshop, ICLR 2026
	Poster , “BIOGEN: Evidence-Grounded Multi-Agent Reasoning for Transcriptomic Interpretation in AMR,” RECOMB Satellite Workshop on Regulatory & Systems Genomics, Greece 2026
SELECTED COLLABORATIONS	Prof. Ser-Nam Lim (Associate Professor of Computer Science, UCF; formerly Senior Research Scientist Manager at Facebook AI / Meta, 2018–2023). Supervises my work on correlated failure in LLM-judge consensus. Prof. Björn Schuller (TU München and Imperial College London; CSO, audeERING): affective computing and computer audition. Prof. Ravid Shwartz-Ziv (New York University): information-theoretic deep learning. Prof. Rajib Rana (University of Southern Queensland): speech and health AI. Dr. Pegah Khosravi (Institute for Artificial Intelligence, UCF): foundation models for healthcare. Dr. Andrea Rivera-Sepulveda (Pediatrics, Nemours Children’s Health) and Dr. Niall Higgins (Queensland University of Technology): clinical evaluation and validation.
HONORS & AWARDS	• Graduate Presentation Fellowship, University of Central Florida, 2025. Selective award funding research presentation at the Molecular Machine Learning Conference, MIT Jameel Clinic. • Finalist, Dr. Pradeep P. Thevannoor Innovation Awards, India, 2019. Selected to the final round through international competition. [news] • Earlier awards: Winner, IEEE SS12 Innovation Challenge & Maker Fair (2018) for the Kidnap Prevention Mobile App; Divisional Champion & Global Nominee, NASA Space Apps Challenge (2017).
PROFESSIONAL SERVICE	Journal Peer Reviewer, 2024–Present. <i>Scientific Reports</i> (Nature Portfolio), <i>Computers in Biology and Medicine</i> (Elsevier), <i>Computer Methods in Biomechanics and Biomedical Engineering</i> (Taylor & Francis), <i>PLOS ONE</i>, <i>INQUIRY</i> (SAGE), <i>Digital Health</i> (SAGE), and <i>International Journal of Computational Intelligence Systems</i> (Springer). Workshop Peer Reviewer. ICLR Trustworthy AI Workshop, 2026.
TEACHING	Graduate Teaching Assistant, UCF Aug 2025 – Present STA 3032: Probability and Statistics for Engineers. Graduate Teaching Assistant, Mississippi State University Jan 2024 – May 2025 Five graduate courses spanning software testing, computer forensics, machine learning, and artificial intelligence.
	TECHNICAL SKILLS LLMs & alignment : RLHF, DPO, SFT; agentic systems; inference-time steering of frozen models; continual pretraining; multimodal modeling. Trustworthy AI : uncertainty quantification, calibration, Bayesian inference; LLM evaluation and benchmarking; runtime safety monitoring; adversarial robustness; graph neural networks. Tooling : PyTorch, Hugging Face Transformers, scikit-learn, LoRA / QLoRA, multi-GPU training, FAISS, RAG pipelines, Python, R, SQL, Docker, Git, Azure, GCP.